

Methods	Suspense			Novelty			Enjoyment			Logical Sense			Naturalness		
	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%
Ours v.s. Baseline	84.9**	11.8	3.3	76.6**	13.6	9.8	67.9**	19.4	12.7	49.2*	27.8	23.0	25.3	18.6	56.1**
Ours v.s. Ablation #1	75.0**	19.2	5.8	63.1**	16.5	20.4	59.9**	17.2	22.9	37.8	27.1	35.1	31.0	29.9	39.1
Ablation #1 v.s. Baseline	79.8**	12.9	7.3	68.9**	11.3	19.8	71.7**	12.1	16.2	45.3	30.9	23.8	33.2	28.6	38.2
Ours v.s. Ablation #2	80.4**	9.9	9.7	66.0**	10.3	23.7	68.6**	16.6	14.8	39.4	32.4	28.2	22.8	24.0	53.2**
Ablation #2 v.s. Baseline	72.3**	12.7	15.0	74.3**	9.9	15.8	63.1**	15.3	21.6	29.6	22.3	48.1*	30.9	25.4	43.7

Table 3: Human evaluation results for both the comparison against the ChatGPT baseline and the ablation studies, showing the percentage of times stories generated from each method are preferred for five evaluation metrics. “Win” means the method on the left is preferred. ** indicates results are significant at $p < 0.01$ (* at $p < 0.05$) confidence level using a Wilcoxon sign test.

Methods	Suspense			Novelty			Enjoyment			Logical Sense			Naturalness		
	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%
Ours v.s. Baseline	73.9**	15.7	10.4	66.2**	9.6	24.2	59.8**	18.3	21.9	42.6	23.6	33.8	25.1	19.9	55.0**

Table 4: Human evaluation results for the comparison against the Llama 2 baseline, showing the percentage of times stories generated from each method are preferred for five evaluation metrics. “Win” means the method on the left is preferred. ** indicates results are significant at $p < 0.01$ confidence level using a Wilcoxon sign test.

human perception of narrative suspense, and effectively produces more novel and enjoyable stories. Our method also achieves improvement in the logical sense of stories. We attribute this to our method’s outline planning component which can make the narratives more rational and reasonable. As our method and the baseline method both use ChatGPT, the naturalness of our stories stays at the same level as the baseline’s.

To further validate the effectiveness of our method across different LLMs, we perform an additional experiment on Llama 2 (Touvron et al., 2023) (Llama-2-13b-chat). We follow the same experimental setup as introduced before, except changing the LLM from ChatGPT to Llama 2. Results presented in Table 4 show that the Llama 2 augmented with our method can generate more suspenseful, more novel, and more enjoyable stories compared to a vanilla Llama 2. These results echo the main results reported in Table 3 and therefore prove that our method is stable and transferrable across different LLMs.

Since the stories generated by our method are long-form (see Appendix C for an example), we include Re3 (Yang et al., 2022b), a commonly used long-form story generation technique, as an additional baseline to compare with. We condition both Re3 and our method on the exact same background information to generate full stories. Results are presented in Table 5, showing that our method can generate more suspenseful, more novel, and more enjoyable stories compared to Re3.

4.2 Ablation Studies

To fully investigate the effectiveness of our method and confirm the necessity of two components: outline planning (§3.2) and detail elaboration (§3.3), we perform two corresponding ablation studies. First, to study the necessity of detail elaboration, we ablate our method (denoted as Ablation #1) by replacing the detail elaboration component with a straightforward prompt as presented in Table 6. Second, to study the necessity of outline planning, based on Ablation #1 we ablate our method (denoted as Ablation #2) by further replacing the outline planning component with a similarly straightforward prompt (also listed in Table 6). We recruited 90 participants and followed the evaluation setup of subsection 4.1. Participants’ preferences presented in Table 3 with a fair average inter-annotator agreement show that both outline planning (§3.2) and detail elaboration (§3.3) are necessary for generating novel and enjoyable suspenseful stories.

4.3 Examination of Story Outline

After confirming the effectiveness of our method and the necessity of individual method components, we need to further examine the story outline generated in subsection 3.2 to determine if our method’s internal workflow is running properly as we expected. We asked (1) whether the actions of the protagonist are relevant to the goal, (2) whether a plausible reason is given for the protagonist’s failure, and (3) whether each of the protagonist’s actions, after each failure, is perceived as decreas-

Methods	Suspense			Novelty			Enjoyment			Logical Sense			Naturalness		
	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%
Ours v.s. Baseline	81.1**	6.8	12.1	84.7**	11.3	4.0	70.8**	11.9	17.3	33.1	25.7	41.2	28.4	12.0	59.6**

Table 5: Human evaluation results for the comparison against the Re3 baseline, showing the percentage of times stories generated from each method are preferred for five evaluation metrics. “Win” means the method on the left is preferred. ** indicates results are significant at $p < 0.01$ confidence level using a Wilcoxon sign test.

Ablation #1’s Prompt
Story Background: \$name_occupation. \$goal. Action #1: \$action1 Reason #1: \$reason1 Action #2: \$action2 Reason #2: \$reason2 Action #3: \$action3 Story Summary: Aiming to achieve the goal: \$goal, the protagonist first tries to take Action #1, but fails due to Reason #1; then tries to take Action #2, but fails due to Reason #2; finally tries to take Action #3 and this time the protagonist successfully achieves the goal. Write a full suspenseful story based on the story summary.
Ablation #2’s Prompt
Story Background: \$name_occupation. \$goal. Story Summary: Aiming to achieve the goal: \$goal, the protagonist first tries to take a first action, but fails due to a reason; then tries to take another new action, but fails due to another reason; finally tries to take a new action and this time the protagonist successfully achieves the goal. Write a full suspenseful story based on the story summary.

Table 6: The prompts we use in ablation studies. All the variables shaped as \$variable are obtained in [subsection 3.1](#) and [subsection 3.2](#), as reflected in [Table 1](#) and [Table 2](#).

ing in likelihood.

For each aspect, we recruited 30 participants to examine the goal, actions, and reasons for failure of 30 randomly selected stories generated by our method. On the question of **protagonist action related to goal**, participants reported that protagonist actions were relevant to the goal 96.5% of the time. Regarding **plausible reasons for failure**, 89.8% of the time. We find that in the vast majority of cases, the action is relevant to the goal and the reason is plausible for the protagonist’s failure. On the question of **decreasing likelihood**, 55.1% of participants indicated that they thought the likelihood of actions was going down, 13.9% indicated that they thought the likelihood was going up, and 31.0% thought the likelihood stayed the same. More than half of participants perceived declining likelihood trends, and a clear plurality felt that likelihoods were decreasing or staying the

same. These results all indicate that the internal workflow of our method is running properly as we expected.

5 Factors Affecting Suspense Perception

Given the success of our method in creating suspenseful stories, we recognized a unique opportunity to contribute to a more fine-grained understanding of suspense. Prior work from psychology used hand-crafted stories from literature and thus did not have the means to run controlled studies with a large number of stories with fine-grained interventions. We conducted a study to investigate the timing and ordering of knowledge revealed to the reader, as well as the relationship between empathy for the protagonist and suspense. We specifically investigate how the following affect suspense perception: (1) clues that set up the failure; (2) information revelation timing (before vs after); (3) reader’s empathy.

Clue setup: we recruited 90 participants to ask for their preferences regarding the suspense level of 30 randomly selected pairs of stories. For each pair of stories, which is generated with and without clue setup (detailed in [subsection 3.3](#)) respectively, we ask participants which one of the two stories makes them feel a higher level of suspense. 57.9% of participants reported higher suspense when reading stories with clues, whereas 10.9% of participants reported higher suspense when reading stories without clues (31.2% were a tie). These results are significant at $p < 0.05$ confidence level using a Wilcoxon sign test. Thus we conclude that clues of impending failure are a factor that increases reader perceptions of suspense.

Information revelation timing: we recruited 90 participants to ask for their preferences regarding the suspense level of 30 randomly selected pairs of stories. For each pair of stories, which is generated by revealing the information *beforehand* and *after the fact* (detailed in [subsection 3.3](#)) respectively, we ask participants which one of the two stories makes them feel a higher level of suspense. 42.1% of participants reported higher suspense when read-

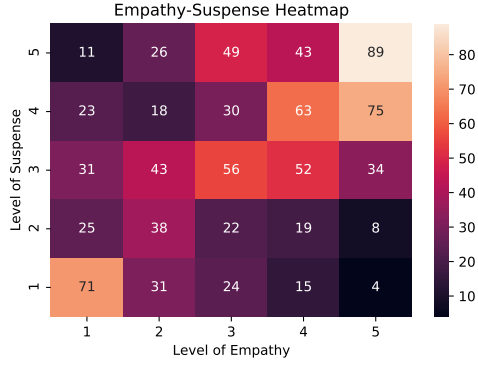


Figure 2: Human evaluation results for the investigation on reader’s empathy as visualized in a heatmap. The numbers in the heatmap indicate the counts of cases in which human participants feel the corresponding levels of empathy and suspense after reading a suspenseful story.

ing a story that explained the impending failure beforehand, and 36.3% of participants reported higher suspense when the impending failure was explained after the fact (21.6% tied). This indicates that it is slightly more effective in facilitating reader suspense perceptions by explaining how the protagonist will fail beforehand than after the fact. These results are not statistically significant.

Reader empathy: we recruited 30 participants to ask for both the level of empathy for the protagonist and the level of suspense they feel (level=1/2/3/4/5) after reading 30 randomly selected suspenseful stories generated by our method. Results are presented in Figure 2 as a heatmap which shows that a high level of perceived suspense is usually associated with a high level of reader’s empathy. However, there does exist a noticeable portion of participants who exhibit a very low level of perceived suspense even when they are reading suspenseful stories, associated with a very low level of empathy. One possible explanation for this phenomenon is methodological. Not a lot of character setup is given in the experiments and empathy may not be triggered strongly in some participants, who then also do not perceive suspense.

6 Conclusions

In this paper, we propose a theory-grounded *iterative-prompting-based planning* approach to generating suspenseful stories. To the best of our knowledge, this is the first attempt at suspenseful story generation with LLMs. Through extensive human evaluations, we prove that our method is more

capable of producing suspenseful stories than the baseline method. Our in-depth analysis of factors that affect suspense perception is also informative for story generation researchers and practitioners.

7 Limitations

Our system is developed for suspenseful story generation in English and our method has not been tested in other languages. The method is grounded in theory based on analysis of stories from a Western storytelling tradition. We believe that our method intuition is directly transferrable to other language scenarios, but to develop a fully functional non-English suspenseful story generation system, much additional effort must be required, especially for the necessary prompt engineering and human evaluations.

8 Ethical Considerations

This work uses human participants for evaluation. All procedures in this work have been approved by our institution’s Institutional Review Board (IRB). We provide the following statement on the first page of all our surveys, to make the human participants aware of the potential risk: “This survey may contain sensitive, distressing, or potentially triggering content. Please proceed only if you feel comfortable and are prepared to engage with thriller stories.”.

We ensure the human participants are fairly compensated by paying them at (and sometimes above) the hourly rate recommended by Prolific. We reasonably estimate the survey completion time to be two or three times (depending on the survey’s difficulty level we perceive) the time we have spent taking the survey.

References

- Prithviraj Ammanabrolu, Wesley Cheung, William Broniec, and Mark O Riedl. 2021. Automated storytelling via causal, commonsense plot ordering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 5859–5867.
- Edward Branigan. 2013. *Narrative comprehension and film*. Routledge.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.